

John Vervaeke ~ Active Inference Insights 003 ~ Relevance Realisation, Predictive Processing, Flow

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okay and we're away um welcome everybody to the third episode of active inference insights brought to you by the active inference Institute I'm your host darus pvy Wayne and today I am absolutely thrilled to be able to speak to John BVI John is an award-winning professor of psychology cognitive science and Buddhist philosophy at the University of Toronto he is also the presenter of the renowned YouTube series Awakening from the meaning crisis as well as the newer after Socrates his work focuses on 4E cognitive science which holds that cognition is embodied embedded enacted and extended beyond the brain in particular John explores relevance realization our adaptive ability to zero in on salian information in a world of near infinite complexity last year he Mark Miller and Brett Anderson wrote the paper predictive processing and relevance realization exploring convergent solutions to the frame problem which proposes that tradeoffs uh trade-offs in Precision waiting a key second order dynamic in predictive processing hierarchies are at the heart of our ability to be intelligently ignorant and that alignment of predicted processing and relevance realization is exactly what we're going to be talking about today John welcome to the show thank you so much for joining us it's an absolute treat thank you Darius uh a great pleasure um I'm uh hoping that Mark does make it on in December uh that'll be great yeah absolutely but in his absence because Mark was meant to be here today but it's uh unfort he's not unfortunately able to make it um perhaps so this podcast is acting as a kind of primer an introduction for our audience to the critical themes in cognitive science and psychology and biology as well as maths and physics as they are right now um so I think it would be a worthwhile uh worthwhile place to start if you could just unpick what relevance realization is and then we'll come to how it aligns with predictive processing sure uh so relevance realization is um sort of inverts the way Common Sense Works common sense is there's a lot that is obvious to us and we know what it's obvious what we should pay attention to we're sometimes mistaken but it's obvious what we should be remembering and what we should be doing and that's all obvious and we just run from that um

our job as cognitive scientists is to explain how the brain generates that obviousness that salience Landscapes that makes the right things stand out as relevant to us so that we can do what is astonishing solve a wide variety of problems and a wide variety of domains um you're a general Problem Solver which is just astonishing I mean you can learn Albanian history you could learn about swimming or rock climbing you can take up the study of dinosaurs in the Jurassic period like it's just you know and it's just amazing and astonishing and there's good empirical evidence you have something like a general intelligence there seems to be a general capacity there's individual variation and talent some people are better at learning this domain than other domains but there way back from Spearman we know there's some sort of General ability uh I'm going to propose that the general ability are two interlocking things the anticipation and relevance realization what I mean by a general ability is the these are kind of meta problems these are the two big problems you have to solve when you're doing any other specific problem solving and what makes relevance realization so hard generating that obviousness so mysterious actually that's what Scott atran says he he says that what science does is it makes what common sense to take takes to be obvious mysterious here's a table in front of me but physics says well what is that really and mysterious and it's made out of quarks and all this sort of stuff it's the same sort of thing here because when you think about it um outside of common sense you realize that there's just an overwhelming amount of information that's constantly as you said dynamically changing in the environment there's an overwhelming amount of information in your long-term memory that's constantly changing and being readjusted during to to the reconstructive nature of memory there is an overwhelming number of combinations of actions you can perform see sequences of actions and yet you ignore that overwhelming almost all of that overwhelming information and zero in on the relevant information so that you are oriented right in the world finding things obvious Salient standing out and you're and and it's and it's not just a single thing there's like a salience landscape some things stand out more than others some things are more foreground background you have all this happening so that you are capable of solving so many problem problems in a messy complex World in which there's constant novelty because of an emergent phenomena and trying to give this ability to machines has been overwhelmingly uh difficult and one of the hard hard problems we're actually bumping up right into it now as we've finally taken up the AGI project the project of trying to create artificial general intelligence excellent um it may be worth digging a little bit deeper into that Exposition um because people might be thinking well don't we just have a module for relevance or salience don't we just kind of we're

born and we know oh um I should not fall off cliffs and I should but actually what happened yesterday might be quite irrelevant you have a wonderful um explanation in your sort of keynote paper explicating relevance realization where you say that leads to an infinite regress um perhaps you could explain that argument so the magic module not to be confused with predictive processings reported problem of the magic modulator so let's keep those two distinct from each other but the mag the magic module is well I have a I have a relevance realization thing in my head that just does it and then that just and then I can just say well how does it do it and the mistake people can make is well Evolution made it and that's right Evolution tells me how it got here that's not telling me how it functions Evolution tells me how my eye got here and how my ear got here and that's not telling me how my eye and my ear function I'm trying to take What's called the design stance how could tell me what I need to do to give that module the ability to realize relevance you you all you've done is said it has that ability but it faces the problem of okay now it has to know what to pay attention to and so on and so forth now what you might say is well Evolution totally prepared us for it the the problem with that is that doesn't work uh Evolution can only pick up on things that are longterm invariant and that make an ongoing continuous um difference to your reproductive uh status right and that's not how relevance Works relevance is really fast and changing if I say left big toe it suddenly becomes relevant to you and it wasn't relevant a minute ago and what's the darwinian difference there um and there's nothing that's intrinsically relevant well my own life is intrinsically relevant is it you'll sacrifice your life for your kids well my life and my kids right under all circumstances you know what if you know saving your kid means that 10 million people oh well and so on and we get into all these philosophical arguments because we realize no no there isn't any hard fast thing that is always relevant relevance is not something for which we can generate a scientific theory it's not stable it's not intrinsic to the phenomena and it there's nothing other than being relevant there's nothing that all the events or objects or feelings have in common like what is it that what do they all have that makes them other than saying well they're you'll give me synonyms they're important to me or they help me solve my problems yes that's exactly it but how but how exactly perfect and it's it's so nice to see how the um apparently disparate strands of cognitive science so gibsonian affordances for cognitive science predictive proing processing and act of inference a line at this path which is that everything is dynamical everything is about this Mutual unfolding there is no rifed static um existence it's what's relevant to you might not be relevant to me and there are these Dynamic relationships which govern that um it's yeah so that's that's a that's a great place to

start I think let's now go into uh the solution if there is such such a solution and let's start with opponent processing so this is a term that you use frequently and to a Layman audience it's not clear exactly what it means so what what is opponent processing and how does it help shed light on this complex problem sure um and maybe along the way I can point out how relevance realization really gives some specific tee to claims of embodiment from for eoa because I I'd like the point you made and maybe we don't have to do it right now but I'd like to come back to it how relevance realization is kind of a glue that glues together predictive processing and for e cogi in powerful way and makes them all including itself mutually stronger from that integration uh but yeah the opponent processing so let's just note something at many different levels of analysis in your biology you will find opponent processing at work work I'll take one that's very easy to explain and people have a ready experience of it uh so we are all constantly mostly unconsciously although it's affected by conscious factors we're constantly recalibrating and adjusting I dare I say it we're even evolving our level of metabolic arousal I don't mean just sexual arousal don't be freyed here I mean arousal how much how much how sort of activated are you how sort of energized are you um and the problem is there isn't isn't some State you homeostatic State you're looking for with that because if there's a tiger in the room I need to go to maximal right arousal uh uh and if I'm going to sleep I need to go to minimal arousal and I can't just be sort of Canadian and keep sort of average arousal at all times because then I don't fall asleep and I get killed by tigers and so it constantly has to be like we're talking about has to constantly readjusting so what what has evolved in your biology is your you're autonomic meaning self-governing it's a self-organizing system like you said it's a dynamical entity your autonomic nervous system and what it does is it couples together two subsystems that are have opposite biases from each other your so your sympathetic system is biased to I I'll speak anthropomorphically just because it speeds things up your sympathetic system is biased to seeing as much of it can of the world as threat or opportunity and arousing you as much as it can and your parasympathetic system is biased the opposite way it's biased to seeing as much of the world as secure safe A place where you can rest and recover and then these subsystems are not independent from each other they're locked together and they're they are ALS they're continually trying to shut each other off and they're constantly competing that way but they're cooperatively competing it's not adversarial they're cooperatively competing and what happens is the constant tradeoff between those those proc between those processes constantly right in very Dynamic manner constantly recalibrates your level of arousal to the world and that's opponent processing and what you find is you got that

you know there's other there's opponent processing between your focal vision and your peripheral vision there's opponent processing plausibly between your left and right hemispheres the number of these things like uh uh is huge and I say at many different levels of analysis um and so um I think opponent processing is a clue uh to how relevance realization is actually undertaken by your embodied uh cognition wonderful and immediately to anyone interested in Act of inference or um its manifestation in continuous State spaces as predictive processing it what we'll be screaming out is this notion of an attractor set for our homeostatic equilibrium which Carl has spoken about at length obviously and and as has many other researchers but this idea that what this we are we suffer um but we also uh embody in a kind of positive way this itinerancy we we never we're never stuck in a single um mode we we are owed with the capacity to go beyond our homeostasis and return back to it but we're always in a sense being drawn without ever residing ever without ever having stopped at that homeostatic set point um so let's now fold in active inference to the picture what does it provide relevance realization um and the formulation of relevance realization that couldn't be done purely with for cognitive science so I mean I'm going to talk about it in terms of predictive processing because that's the one that's the formulation of it that lines up most cleanly uh with the theoretical integration uh as you said there's there's clear derivation relations between active inference and predictive processing and also between them and the basian uh math but the basian math if we were to actually strictly apply it would be computationally intractable so we're doing some approximation function so I'm just going to take it that everybody sort of knows that I'm playing fair with this right I'm not I'm not trying to be Dodge you here um and so um relevance realization is grounded in uh the idea of problem solving and problem solving actually assumes a fundamental thing which again is so obvious to us which is the state you're in and the state you want to be in are not the same state that's the defining feature of a problem if I'm in a state I want to be in I want to be sitting in this chair I am I don't have a problem um right and now what that means is you're all you're you're immediately into something very interesting the organism is trying to actually predict a possible state in the world and prepare itself alter so it's an agent it doesn't just behave it Alters Its Behavior to alter the con the states in the world right and so it is trying to I would say predictively prepare for the world it's trying to predict the world but prepare itself for that world but also prepare the world so it's more likely to come out in the prediction that it's seeks right and so I I put those two together um and I I talk about anticipation um and so you're whenever you're problem solving you're anticipating a goal meeting that prediction and

preparation now for EOG site doesn't directly talk about that as clear as it needs to it talks about coupling and it talks about affordances and I think there's a deep connection between opponent processing optimal gripping and affordances and maybe we can that at some point uh but typically one of the things where uh 40 kogai has some challenges is in more distal relations uh to the environment because it tends to rely primarily on coupling and you know this is a long-standing critique that one of my colleagues Brian Kwell Smith at the University of Toronto because that is where all knowledge is flowing from um right uh has made he said you know uh you can be predictively coupled to things that you're in direct causal contact with but you know uh um uh sorry dynamically coupled not predictively coupled my my apologies uh but you how do you do things that are much more distal and for me that that is key and here's why darus I think in I think when we evaluate even intuitively um so I'm not using that as an authority I'm just showing how readily this works for us um when we evaluate an organism for its intelligence we tend to to it in two interlocking ways is we do it in terms of I think how how well does it zero in on irrelevant information how well does it pay attention to what it needs we look at an animal we go wow that's really good you notice how it's noting subtle differences but we also evaluate the intelligence of an organism and Michael Levan talks about this with this cognitive light cone idea is How Deeply into the world they can anticipate and I don't mean just spatial temporally I also mean it Mally possibility like when I talk to my cat and I say you know you know where's your toy the cat looks at me and I go oh well right I say to my puppy where's your ball and she goes into the other room looks for the ball finds it under the couch and brings it all the way back to me I go wow you're really smart right and so we get this because we we know how just Moment by moment that adaptive ability is built on what how dist can we pursue goals we tend to evaluate people wow you pursued a long-term goal and you brought it about so that's what's missing and I think that's what what's really afforded now I want to make one more point and then I'll I'll shut up so you can you can ask me another question which is these two issues anticipating more deeply and remember I don't just mean spatial temporally I mean Mally and relevance realization are deeply interconnected the more you anticipate the more the problem of relevance realization goes up exponentially and so these two things right they I I would argue because they're interlocking they have to be solved together and I think that's why we use both of them as evaluations of the intelligence of an organism okay I'd like to backtrack to that second argument after this after what I say here it um yeah it it strikes me that what predictive processing can give perhaps where for cognitive science doesn't but I'm happy to be schooled

on this um and I know verel and Auto pesis in some sense pre- shadowed um and prefigured what I'm about to say is that it kind of gives a ground to our action and to our perception which is that we have to have a model of the world and a model of ourselves which is not only descriptively viable but also normatively viable yes in so far as it's all like it's very important for me to be able to predict the world but as you said we aren't just at the behest of the world's Dynamics we can change the world according to our preferences exactly and I think what predictive processing gives us is although although to be fair I think Carl might say that this is a bit of an overshoot is a is a Telos is a fundamental attractor set to which our perception and action is is governed so I think that's kind of the way that I see the the added benefit of this convergence between um predictive processing and um relevance realization what I was also going to say there is that it's fantastic that you're mentioning this kind of deep temporal modeling because for my eyes that's really where a lot of the the work is being done right now which is that selfhood Consciousness even perhaps space itself is Downstream on the fact that we can Downstream on the degree to which we can model the slower Dynamics and the faster Dynamics in a generative hierarchy so that's I think that's I I'm in complete agree that I think that's an extension of the argument I made uh yeah and so what I wanted to pick up there is you said if I heard correctly that the more the the deeper your temporal model the more you can model slower fluctuations in the environment the more critical relevance realization becomes the higher the stakes in some sense yes yes very much I'm interested in unpicking that argument because for my eyes there there are plenty of things out there that don't have deep temporal models so um I don't a virus for example would just be reacting to very uh Co brain features of the environment but for them relevance realization at least to my eyes appears as critical as it would for a human it it is it is and and so I think it's how how can I put it I think it's always a demanding problem I'm so I'm not I'm not denying that even uh the uh the parium has to uh do salience land cap aping it has to I'm going to use a word very neutrally here I'm not conting anything about Consciousness but it has to recognize this molecule as food and that molecule as poison and swim reliably towards the one and away from the other I'm not I so I think the problem is always there as I was trying to indicate as soon as you have problems as soon as your goal states are distinct and I think the space opens up very quickly exponentially I just meant that it exponentially gets worse right this this is is why you see uh across species hyperbolic discounting temporal discounting because that's a that's a huge relevance realization machine it's about salience discounting right and the point of that is is because as you open into the future the number of possibilities goes up exponentially and so

you need to have this attenuation function so you don't get overwhelmed by the possibilities as you extend your cognition into the future that's why we have hyperbolic discounting and we pay a huge price for it it makes us procrastinate makes us difficult to give to pursue our long-term goals so there's even a trade-off relationship in there but I won't get into it but that's an example there's a good reason why we must have something like temporal discounting across species that's what Angley showed is because the relevance realization issue just it gets even worse it starts out bad and gets even worse right wonderful and yeah I mean a lot of your work we we we'll come to affordances now because a lot of your work is centered on the notion of affordances yeah affordances are the possibilities for Action afforded actually circular possibilities for Action granted by the environment I mean afforded was a verb and then Gibson made now um how much has the predictive processing framework supplemented your understanding of affordances and perhaps the way that what we perceive in the world is not necessarily a feature list of objects but um they are fundamentally affordances I see my glass of water as gri rather than being glass this Dimension and I only ask that because as I said as as conscious animate cognizers we are blessed with the capacity to change the world in accordance with our priors which makes action really the fundamental currency yeah for autoparesis for self-organization so does it has predictive processing ramped up the uh importance that you give to gibsonian affordances um I don't know if it ramps up because I I always thought uh I I literally learned this stuff from John Kennedy uh when Gibson's greatest prot so from very early on I was very much uh impressed by this um and it it occurred to me that relevance realization um the realization of relevance and I mean realization in both senses actualizing a possibility and becoming aware of it or at least detecting it in some fashion um that it it's a prototypical kind of affordance the thing about affordances is they're not found in the object or in the organism the grip ability of the glass is not in the glass can't be gripped by paramuseum and it's not in my uh you know in my hand my hand can't grasp Africa or the sun right it's it's in a fitted relation between them and that's exactly what relevance is it's a fitted relationship between so I always before I I came into a a deep dialogue with with predictive processing I already saw a deep relationship between relevance realization and affordances and I think um part of my work has been to really explore the Deep ontological significance of that that you have a category that sort of Falls outside this was Gibson's intent too this is what 4 Eide is really talking about Falls outside are our belief that we have a complete exhaustive dichotomy between the subjective and the objective between the inner world and the outer world and so there's this other important I call it the transcor this betweenness this

connectedness you see in adaptivity you see in relevance but you also see it like you said now what do what do I think the predictive processing did I I I it's if you'll allow me I want to talk this just I want to talk back and forth symmetrical not just linear so I think what the predictive processing does is emphasize the need to try and explain rather than just I don't want to be cruel so I'm being a little bit overs simplistic I'm doing that just for Speed sure but generally it's like yeah but how do the affordances get actualized in the 4E coln like there's a lot of affordances all the time and what I'm interested is when how do those affordances become well Salient and obvious to me so they become binding on my sensory motor behavior and that was always something of uh you know not that you know there's Redfield and others who were doing some work on it but to typically they start to invoke predictive processing to try and explain that mechanism of like okay there's a there's a huge affordance network but I'm not like how do I here we go again how do I select and actualize the ones that I'm actually going to that actually go into my salian landscaping and I think predictive processing especially with the notion of precision waiting does a good job of that now what I think relevance realization helps to do to predictive processing a bunch of other but one thing is predictive processing basically there's subtlety to this and I know there's lots of math but just to keep going right that you know well what attention is is the Precision waiting it's this meta function and I think that's a very good argument but there but there's a conceptual analysis that's been needed but it's like but what is saliens and one of the arguments I can make is well what the Precision waiting is doing is relevance realization and relevance realization uh when relevance realization re does a higher order finding relevant of an affordance that has been generated by lower order relevance realization that's when something becomes Salient to us and so we can actually give a conceptual exclamation that goes with the theoretical identity claim in predictive processing the Precision waiting is attention and what it gives you is salance so they can they can unpack each other I think what another thing that happens is relevance realization is deeply connected to for Eide because relevance is grounded I would argue in auto pois um relevance relevance is always relevant to an organism right and so that means relevance realization isn't called calculation it's how the organism cares about some information rather than all the other information because it is constantly taking care of itself there are literally and I mean this physically things that matter to it and things that it must import into itself important things and some of that's also information and so by that if you if you can if if you can glue for EOG sign predictive processing together I think you can answer some of the questions maybe even challenges that some people in for E cogi are making a

predictive processing and saying well it's not really connected to embodiment I think no no if you show a deep theoretic in theoretical integration of predictive processing relevance realization and then relevance realization and autois then you've really strongly glued predictive processing and for E cogi together so that's some of the that's some of the important theoretical work that can be done yeah absolutely I I would love to jump into um the critiques that certain fory um cognitive research uh cognition researchers have leveraged at predicted processing but let's um let's leave that for a second this is exactly what I was talking about in terms of Telos um which is that exactly this this notion of care and and when I spoke to ma alaras in an active inference research um last week we kind of actually ended up concluding this is almost the hiigaran sense of care that on our existence can I can I just interrupt there yeah right that hiigaran sense of care was exactly what I was in invoking and I right and I was deeply influenced by dfus and the frame problem got this from haiger right so the the connection you just drew is very I I that yeah I I explicitly argue for that excellent so yeah I mean I personally think um that the hiig hiigaran draus Merlo Ponty lineage needs to be integrated more into yes active inference and I'm speaking to Dr Marilyn stendera who has done work on hyar and auto paresa so that will be really that would be really fun in the upcoming weeks but this is exactly what I was talking about in terms of Telos which is that taking a stance on your own existence can be computationally modeled as having these kind of high Precision priors yes so our homeostatic set point will be different from snakes and that actually gives a really solid explication of why we act differently in certain context than snakes um and so that was my kind but then that said I will have to make clear that Carl is in himself although he has spoken about existential imperative says that there actually is no imperative all that you're really saying is if a thing like us is to persist over time this is what it needs to do and it's just a it's just cast as a minimization of free energy so I think there's an interesting question there that maybe you could have a go out which is to what degree do you see this as a genuine imperative um versus just what things do to self-organize over time yeah and so I mean and and uh Carl's no SLO so he's welcome I I I sometimes don't like it when people who you know are physicists or something start commenting on philosophical uh normative questions uh with an authority they don't properly possess uh but I know I mean Carl's a great theorist and um and I know he's philosophically educated um I have not yet to talk to him but I hope I do get to do so we can do ahead so um yeah I mean so I'm gonna argue later if I get a chance that what relevance realization doing is strongly analogous uh obviously a very different time scale uh to what evolution does about constantly uh uh redesigning the

adapted fitted of the of organisms to the environment and allowing organisms to fit their environments to them because as you keep emphasizing and I keep agreeing with it's not passive it's not right the organism is shaping the environment as the environment is shaping the organism Niche construction and all that sort of thing um and I think that's all right and I and then there again that's a very clear connection to for Coogi again but this deeper question I do want to do it I want to pause and I want to slow down because it's it is a really important question uh because it gets us beyond what we might call scientific explanations of how and into properly but not useless philosophical reflection of sort of why and why does this matter to me because as you've mentioned I'm deeply concerned with you know issues of meaning in life this this this largely metaphorical nebulous notion that humans seek meaningful lives lives that are worth living even given all of our failures and our faults and our flaws and our foolishness and our frustration right and pretty clear empirical evidence that it's not reducible uh to just subjective well-being or pleasure and pretty good conceptual philosophical argument because it's that's where it's relevant that it's not reducible to just living a moral existence you could live a very moral existence in which you're sort of experiencing the pleasure of food and other things um and and and you you know certain stability in your environment but you can be very very lonely you could be very lonely um so and what that loneliness points to I'm just making an intuitive gesture that's not a tight argument I have tight arguments but the the gesture is people are seeking a kind of connectedness and now you see where I'm getting to relevance realization and what we're talking about this Niche construction the fitted the belonging togetherness this mutually shaping like this is this I argue that that's exactly the connectedness human beings are they're seeking the meaning in life now you can then ask the question and PE and I'm going to play with two terms of phrase here to play with that I think we clearly seek meaning in life uh and I think the evidence for that is growing and it it converges with the psychological work on meaning in life and on what Karen Allen calls belongingness if you don't feel that you belong you're in trouble and you're in trouble across all these measures cognitive emotional social Financial blah like you're in trouble physiologically you're even in trouble um that's why solitary confinement is such a punishment right um so is there a meaning of life to the meaning in life the meaning of life is is there some sort of cosmic Destiny Cosmic order to which we are ultimate which we are ultimately trying to find and here's the thing I don't think I have enough evidence for that I think that there's lots of evidence for meaning in life and if I'm right that it's relevant realization and if that's something very much like Evolution right evolution is constantly

redesigning adaptivity but there isn't a final thing that evolution is aiming towards there isn't a final form of life there isn't some like what we're doing is like the sculptor we're constantly refining and then finally one day we will have the final form of life and um I think this is fundamentally flawed and I think this matters philosophically I don't think previous forms of life were in some in in any kind of moral sense Superior to us so I don't believe in any Nostalgia some the previous things that people found important and relevant and sacred those were the true ones and we just have to get back there nope and I don't believe the Utopia we are working towards the final ultimate thing that we'll all agree for all time is the most relevant most Salient I I don't believe that how this works so I reject nostalgias I reject Utopias the idea that there's a TS can either mean there's a TS to meaning in life which is it's like Carl says it's necessary it's constitutive necessary for living things to have this but does that mean there's any metaphysical necessity to their being living things things that's a different question now here's my weak answer at that one sorry long question but you asked like this is like almost up there with like God question right right I note that I'm in a very difficult situation I can't see any evidence for and I'm not attacking God here as you know I'm very respectful and even appreciative of of religious Frameworks and religious lives but I can't see any evidence for that however if you were to ask me which is a better Universe one with life in it and one without life in it even if you couldn't exist in either one I would say the one with life in it there's some primordial judgment going on there um that has some kind of metaphysical import now it's weak I I I admitted that ahead of time and so I'm giving you a very I'm sorry very long but wishy-washy answer I don't really think I believe anything like the meaning of life but I do think there's a metaphysical import to being connected to reality we find reality being connected to reality inherently valuable for its own sake and we find worlds in which things can realize reality better worlds than ones that don't have that in it that's my answer no it's a wonderful answer and it um it's really helped me and hopefully our audience clarify exactly what I mean by an existential imperative which is that the free energy principle doesn't tell you what you're here for no it tells you if you're here what are you doing like given your phenotype um and given your culture and etc etc so that's a really useful um sort of explanation of exactly what I was going for there um and actually it points to something I said to Carl right at the end of our podcast which at time of recording this came out yesterday so people should definitely check it out because he for sure for sure he's on top form which is I asked him I I mentioned Thomas Nagel so Thomas Nel has this kind of very romantic philosophical notion that life is death is bad because life is fundamentally good and it points to

kind of your fault experiment if we had two worlds one with life and one without life I think we would all intuitively say the one with life is better whether that has any metaphysical import as you say is up for grabs I was going to I'm going to ask sort of um jumping on the back of the initial thing you said which was about connectedness um and belongingness so if we take this from a theoretical or computational stance in active inference what this looks like to me is we embody a model of the world which um we can so we can make fundamentally sound predictions about how the world will unfold and how about we will and how we will act to make that world more um compatible with our preferences now when that was first introduced initially in the active inference literature immediately a response to this came in the form of What's called the dark room problem so this is saw papers in 2011 2012 and and they've been rebutted in multiple ways but the basic idea is why don't humans just seek out dark rooms where maybe they have access to food and water but they don't really see how but in a state where they don't go and explore and they're not curious because what you're seeing there is actually a perfect coupling between your predictions of the world and the world and the way the world is unfolding to your eyes so I was wondering whether you had pondered the dark room problem and where you see exploration and epistemic affordances coming into play here yeah I think that the dark room problem I'm not I I I'm happy with all the other rebuttals um and I don't and I don't know I don't know I don't know them all so I may be stepping on somebody else's toes so if I am whoever you are I yeah um I think this is you know a lockin individualist model of cognition and how we work and I think it is therefore the presupposition well is that no um that's actually I don't agree with that presupposition that our cognition is fundamentally individualistic in that way I think we are you know we are social sociocultural mammals and let me point to one thing you know yes measures of G are very robustly predictive but you know it's also really pred a very powerful way to predicting your behavior your attachment style this is also very robust now that means think about what that actually means and this you know and this goes into the heart of religious Traditions like agapic love right when you have a child you have to invert your relevance Arrow it's not how that being is relevant to you you invert everything around this is agic love and how it's different from erotic love or FIA love you invert everything around right and so that it's how am I relevant to this being how am I relevant to this being and how can I be relevant in a way that turns it into a person turns it into an intelligent rational self-reflective relevance realizer predictive processor meaning maker right and so well first of all where are my attachment relationships in the dark room well there's other people in the dark room soon as

there's other people in the dark room all the problems that you thought you got away from by putting me in a dark room return other people have different goals than me they have different needs they're going to move around differently right we're going to have to decide about when and where and how we gain access blah blah blah all of that immediately unfolds secondly and this is part of for E cogon I think the evidence for extended cognition distributed cognition that we evolve to work in groups and the collective intelligence of that is actually our superpower and other people are making this argument um I think that's clear you standard thing take the Wast and selection task put it an individual person Highly Educated highly intelligent second year you know psychology and a top tier University from the 1960s on you put him in the Wast and selection task and only 10% get it right right you replace that same task with four people who are owed to talk to each other and the success rate goes from 10% to 82% reliably why because we do opponent processing between each other right you have biases different from mine and if we work in opponent processing not adversarial but if I say you're probably a good source for correcting my bias and I'm probably a good source for correcting yours we get the best dynamical self-correction possible and so when you add in the fact of the reality and there's actually evidence empirical and formal for the power of collective intelligence the reality of attachment well then the dark room right becomes filled with other people who are trying to you know band together to solve problems that they can't solve individually and then to me everything that the dark room thought it had as an absurd like a reductio absurdum disappears that would be my response yeah I think eventually the dark room just becomes the world once you start adding the things you need eventually you know eventually it becomes a civilization yeah I love I love all this stuff on distributive cognition I spent the first um six months of this year working at UCL on um experiment finding experimental ground for what's called social Baseline Theory social Baseline theory is this idea that humans like to study a human being in a lab by itself is to study a human being at deficit a human being actually want it's physiological arousal to take one um one factor is at its Baseline when it's with other people and it will FL uate according to stuff like attachment style or the relationship that one has with that person which is really beautiful but inverts our kind of well maybe not common sense but our in institutionalized notion of what it is to be a person which is kind of alone yeah wonderful well I'm going to take this argument a little bit further if if you let me um because this is something that we spoke about a couple of months ago and I think Mark might have a different opinion so it' be cool to sort of unpickle I have this um slightly fuzzy wishy-washy idea that the notion that okay so I always think about a climber

climbing a um rock face or a climbing wall and the climbing wall is a kind of beautiful canonical example of an affordance-based landscape it's rich for the affordances it's got the toe holes the finger holes so in many ways what the um rock in terms of active inference what it allows the rock climber to do is to self-evidence um or or find evidence for its own model about itself um a successful rock climber yeah now that's relatively well established in the literature what isn't established is my kind of again slightly wishy-washy idea which is that the rock climber is offering affordances to the wall and so this this is why I about you know right at the beginning of our conversation I spoke about Mutual coupling so what dynamical systems theory gives us what um autoparesis gives us what act of inference gives us is this idea that um to exist is not to be this kind of rifed itself that um takes an objective stance on the world and actually what it is is this really the fundamental unit of analysis should be this Dynamic Mutual coupling between agent and Arena to use your terminology yeah do you think it's an over I mean I've spoken to Mark about this and he he uses he said it's something like an overextension of the term affordances to say that the human being is offering affordances to the climbing wall I know Mark isn't here to defend himself so again I'm not going to put what but and I will definitely raise this with him again but just for your own ears how does that argument sound to you I like the argument and yeah I don't want to speak you little Mark I'll only speak to the propositions that you spoke on his behalf and I'm very happy to I I mean Mark and I work close together Mark is one of the best people I know Mark's a former student of mine I'm very proud of Mark so um uh but um I actually agree with you now that goes to another connection I make in my work that would probably be a little bit Stranger to the ears of many of your listeners they might say oh I get the why he wants to connect uh you know processing to relevance realization and for coxy but now what you're doing and this is where I do a lot of my what I call my deeper ontological work and this is where the hiigaran stuff really comes to Bear is this is all the work I do on on neoplatonism which may sound like something very Arcane but it again um here's here's the proposal that once we really profoundly accept affordances and once we accept that they are reciprocally realizing I agree with you the forance not only discloses things about me it discloses things about the world and that's a way in which things like heer's notion of Truth is ala as the as an event as the disclosure rather than a static property of our propositions um I think that hang on now as filler argues and I've been arguing for a while and John rusen uh filler's book is called uh Neil play ISM pager in the history of being relation is ontological ground that relationality is actually the the ultimate nature of reality um heiger is actually turning back towards this

because what does that give you you've had if you have that and I think this is a fair way to put it if you have that althetic notion of affordance then you are getting back to the neoplatonic theory of knowing by Conformity knowing by participation and the deeper argument goes something like this um that if the fundamental grammar of your cognition I don't mean the content your content can go all wrong and all but the fundamental grammar the bottom up top down all all this stuff we're talking about if it's not picking up on something fundamental about how reality is structured right then you face a kind of profound solipsism a profound kind of skepticism um and I think that is ultimately leads you into all kinds of performative contradictions that I and I agree with Whitehead they're as devastating as propositional contradiction there's a longer argument there and I could point to please check out some of my talks on on YouTube about this longer argument like pix Stock's uh aspects of Truth where she said you know all the things we used to deize this right we had the analytic synthetic distinction that is broken down under philosophical criticism uh we had the theory fact distinction that is broken down we we had the the like the is a that has broken down like think about relevance in is or an not well blah right it's sort of both right right and so her point is all the things that were used to cleave and then what we tried to do is we tried to make the logical world the thing that sort of stuck the Two Worlds together and that collapsed under goal and other people and so we're back to the idea that we either we either we either go into the lock in cabinet in which we're locked inside of our heads we're somehow getting postcards sent to us we think they might be from an outside world that we think might be out there and we're trying to build it from the postcards and we're doomed that's never going to get us there and so I think if you have an althetic notion of affordance you start to make the argument that how reality is realizing and how we are doing relevance realization are fundamentally participating in the same principles in a profound way um and and I think yeah that I I I I have no doubt that Mark might not not uh want to do this I think Mark would be I would agree with Mark that this is not a direct derivation from sort of classic Frisian presentations of active inference and predicted processing daasi and brain we should settle on a name uh but uh um but I think if you make the connections through deeper into Gibson deeper into fory kogai deeper into heiger you get back to this notion that with the prevalent notion of knowing that when we know something what we're doing is our structural functional organization is identifying with the structural functional organization of the thing and we are we are both participating in that same form that same principle that same grammar and that has all kinds of metaphysical and ontological consequences even uh ethical consequences it means talking about the true the

good and the Beautiful becomes something very relevant together yeah yeah no this is this is exactly where I wanted to go and exactly aligns with the way I'm thinking about the free energy principle at the moment which is I have a skepticism between the internal and the external distinction yes this is something I spoke about with so a classic problem when one starts reading active inference as I spoke about with Carl is whether one would consider the um the agent or the um internal Dynamics is having having a model of the external world or instantiating a model of the external World model excellent excellent this comes back to sort of cyber cybernetic uh formulations what it is to be a good regulator so col Ashby and all of that stuff from the 50s and 60s so I have an inherent skepticism because I would I kind of take the stance actually all that the internal Dynamics are doing is instantiating a particularized form of the external Dynamics and this is very easy once you take out the picture of a homuncular ego just if you look at it in terms of just the particle physics we're just a instantiation of some um of the the physics that defines everything else but we just we just actually seem to have a kind of complexified Mark of blanket or Mark of blanket form of that but there's nothing particularly distinct about that and we'll come to Consciousness because maybe there is um and that's I guess is the big elephant in the room here but it's funny that you mention haiger because haiger to my to my eyes um in my opinion and then obviously draus and Merlo Ponte and the phenomenologists followed haiger really um he's the one who strikes at this um established distinction between a subject and an object yes so the cartesian distinction and then this is where his kind of gripes with SRA came in now my question is is that yes if you read haiger darine the ground of darine is this kind of inter relationality Alf is the disclosure of Truth in an inter interrelational relationship so to speak but there still seems to be some kind of subject who um you know the the hammerer who is hammering away at the nail now from the perspective of the hammerer he's not some rified ego who is objectifying the nail that all that is really being witnessed there from the level of Consciousness is the disclosure of activity but still we can give a description of a hammerer and a nail so to what degree in your thinking do you completely eradicate the subject object distinction um and is there anything that you think predictive processing has to say about that I think uh well oh the second question was don't worry worry about second question I'm happy to think about it with you uh I have put a lot of thought into the first question yeah um and so I mean I think filler is right in his book that our subjective objective divide is our version of the appearance reality distinction and and it's just our version in the ancient world they had a different version so ours is an in-out metaphor how is the inner and the outer connected um and the ancient world

is the problem of the one and the many how is the lower and the upper connected and and it and and what what and once you see that you realize that there's a deeper structural problem that isn't bound to the inner outer or the uh the emergence and the emanation it's it's this it's the connectivity issue and what what what filler argues and there's a lot of other people uh convergent on this is like if you try if you if you go with an Aristotelian metaphysics that what's most real are substances in the Aristotelian sense which are things that can independently exist um then this becomes a deep problem for you and that kind of ontology I would argue and I can make the argument drives you towards a nominalistic um epistemology um and then you get you know Kant's version of it or Kant's version of it where all of the patterns and all the information are just in the mind which means the mind is radically other because it is the only place that information intelligibility actually exists and the world is profoundly absurd in a deep deep way way and I think that leads you into uh well first of all it makes science impossible and it leads you into all kinds of existential and moral dilemmas and if you then take a look at the logic of trying to get you can't actually and this is what filler does very carefully you can't actually get relations out of properties that belong to the relations have to precede the data in a very important way so I think there's a deep at a very deep level I do want to call it into question I do want to challenge that uh that go first of all we it's historical it's cultural we we pretend as if that is the only way in which human beings have related to the world like I said even in the west that's not their problem in the ancient world their problem is the problem of the one and the many it's not the problem of the in order now what does this all ground in I think this grounds in the common problem of the relationship between appearance and reality and here's where I could now say something that aligns with my previous argument about relationality see so and I'm going to borrow a term from uh Heidegger here the hermeneutics of Suspicion the hermeneutics of Suspicion is that appear and we've got it because of Freud and Nietzsche and blah blah blah there's historical reasons and those are valuable critiques by the way but the Hermeneutics of Suspicion says appearances are distorting they're deceptive they're distractive right they're disruptive and what we should do is always question whether or not they are leading us into reality now Marcel Proust has a great argument against that that he gets from I maybe he doesn't get but it's it's in Plato which is wait you're treating real like red like you can just say you can look at an isolated thing and say that's real but real is a comparative term so of course is illusion to say this is an illusion is to say this is an illusion in comparison in relation to something that is more real and that thing is in relation so on and so forth right and what that means is first of all you have to see that those judgments are inherently

relational and secondly you have to call into question the her the the the independence of the hermeneutics of Suspicion the hermeneutics of Suspicion is actually parasitic on the hermeneutics of beauty there right depends on there being things that we agree on by saying that's where appearance discloses reality rather than distorting it right and then as soon as you do that that undercuts the Deep divide because you ultimately make these divisions between the one and the many or the subject and the object by getting a harmonics of Suspicion into the appearance reality distinction that's at least the argument I would make interesting yeah I I like the I like the suspicion of buying Ness uh inherent in all of this um it it speaks to me and I think it speaks to people who interested in active inference and the notion of sort of philosophical vagueness as well um at what point is something real is is fundamentally a comparative term in the web of things that could be considered real um yeah that that that's right yeah I I I'm trying to do this without invoking a binary but um you mentioned earlier 4E cognitive scientists do have their critiques of active inference um and and predictive processing and I I will stick with active inference actually for now and I will explain why later um it's not that interesting so this this was people like Tony KERO or Ed bags um who who are radical in activists and what they're critiquing in a sense is an internalist picture of yes Act of inference so this is some again I'm not putting words into people's mouths but if you read um yob how's um 2016 paper about self- evidencing some of the language May to some people imply that there is a homunculus that has a representation or picture of the world and this to the ears of a inactivist is deeply worrying I would love to as a four cognitive scientist um with obviously vested interest in predicted processing as well I'd love to just hear your take on that debate and um again without striking at binaries and whether you think those critiques are legitimate well I mean depends what you mean by legitimate I mean you know there's also the ones that Evan Thompson made and I hold Evan in very high regard um I mean they're legitimate in that they are they they're wellmade arguments in peer-reviewed journals and so we we have no right to be dismissive of them well if I was to say if I was to say that let's take as an axiom of active inference like this is just um this isn't the case but let's just say for sake of argument that you have some um internalist representations of a statistical model right so you are in a sense you're not just embodying a model you actually have a model so again it comes back to that distinction we're talking about would a inactivist critique of that internalism be um be justified um so the reason why I'm hesitating is this is landing on the swamp of what do we mean by representation uh which is I mean so does the thermostat represent the temperature in the environment and where and everything I'm going to say is

controversial because of the swamp so I want that understood please right I would say no because what's needed is right there there might be some cor there there might be co-variation but that what I think the critiques of the co-variation model of representation locks ultimately model that what did so we had the idea that representations have to be similar to what they represent that's how they represent and then you have all the problems with similarity and Aristotle even could bring that down um and then lock replac it with the covariation model to have a representation is I have something in my head that reliably co-varies with something in the world and that's how it represents it and then the problem with the covariations is they don't give you um the specificity for example of thought right so this is covarying with a bottle with a tool with a man-made object like which is it those are not the same things so not those are not the same ideas but the this is coar and causally with all of that and you have you have the problem then of aspectual isation and and the loan answer of course is we get we get aspects by doing representation but I agree with surl that that's the wrong way around that any representation is inherently aspectual when I represent this as a bottle I'm only picking up on some of its properties in so far as they are relevant to each other in so far as they are relevant to me so representation depends on as aspect realization which depends on relevance realization and this is not what you do with relevance realization you do not represent all the facts judge them to be irrelevant and then zero in so it's ultimately non-representational so what I would say is I I don't know at some level there's something like representations and but if I agree with many arguments that representations are more than co-variation but there's this kind of carness and aspect caring and aspectual isation through them then they depend on relevance realization which grounds in autopolis and is deeply intertwined with predictive processing that would be my argu that would be my response Splendid and I guess I guess what intuitively supports a more representation this picture is consciousness um I think Consciousness is be going to become a recurring theme in this podcast um well it's the Holy Grail right it is it is the Holy Grail um and so so I guess my question here is is that if one was to adopt a radically inactivist view there's nothing in that picture which um needs Consciousness so why couldn't my Mutual coupling with the world just happen with the lights off so this is a argument I guess that's rooted in Charmers um 1995 paper which is you can give me the function but you can't give me the why why are the lights on and so I guess in in other words my my question to is yes that we may have relevance realization um preceding perception or pered or or or grounding perception yeah then why do we have perception in the first place so um again I'm gonna give a gist of an argument that I've spoken about at length

elsewhere and I'm trying to get published on and presented at conferences and so forth so um I don't think you can separate the function and the nature questions which is what trauma's hard problem uh relies on is it yeah yeah you've given me the function but you've tell me nothing about the nature I don't think function works ontologically like that I think function has to be plugged into the ontology and I think as soon as we're talking about the ontology of anything within a living organism we're talking about something functional so I think the questions have to be answered interconnected so let's go let's go back let's say that you give me that um any cognitive agent has to be doing anticipatory relevance realization or it's not going to be a general Problem Solver and then when it's doing that it has to be aspectual using its world right right it's it's not Rea it's not doing all of the all of this but this as a bottle or the molecule as food the molecule right and then if you then start to pay attention to let's look at the Continuum of Consciousness uh let's look at the possibility which I have experienced that's not even the right sentence many people have and this is Foreman's idea of something like the pure Consciousness event and the pure Consciousness event doesn't have well actually I need a distinction here I'm going to claim it doesn't have one type of qualia but it has another type of qualia in it it doesn't have adjectival qualia there's no red there's no blue there's no cap there's no dark there right you're not even conscious of Consciousness you're just conscious yeah right but what what it still has is it still has the adverbial qualia it still has a sense of heess now and the heess is profound presence that's the language that you The nness Eternity the the the integration right the togetherness it's everything is one so all the adverbial qualia are still there and you still have Consciousness that shows that the adjectival qualia are not necessary to Consciousness now I have the other arguments to show they're probably not sufficient because if I give you sort of atomic blips of bless and greenness and they're not bound together and there's not a heess and a nness so you can any way Orient on them I don't think you have Consciousness either right um so I'm not saying that adav Quia don't exist I'm saying that we've held Consciousness Hostage to them yeah and what I'm proposing to you is that what Consciousness is doing is these adverbial qualia which are just right salients which is just relevance realization which is as far as we can tell tied to the best evidence we have it's all controversial about the function of Consciousness it's tied up with working memory and attention which are both doing higher order relance realization we need to seem to need Consciousness for situations of complexity novelty or IL definedness ones that are really demanding on relevance realization so you can make a pretty and there's a lot of convergence actually on what the function of Consciousness is relevance realization and I think if you plug

in relevance realization you can at least get all the adverbial qualia and that I think gives you a lot of what Consciousness actually is cool um yeah let's stay on that verbal qualia because I'd like to integrate what you've just said with a more active inference account of Consciousness which there are and they're they're also diverse but one well I have to ask a question before is does Foreman um say that mine so um feeling that this experience is mine does he is is that one of his verbal qualia that persists in the pure Consciousness event so that that's another Thicket because the debate about this is a raging debate Evan Thompson actually has a good Anthology on this on the self no self debate right right and I think it's I think it's reasonable and complies with most of the evidence which of course is self-report after the fact which is problematic blah blah blah I agree with all of that I'm not dismissing that but that's what we basically have to go on right now um that the ego narrative sense of mine and me and I goes away I whether or not that is a complete loss of the self as as where relevance realization is happening or something like that yeah um that that I'm not I'm not convinced that that second thing is the case okay I think so if you would allow me and this is tortured distinction a distinction between the ego and the self I think these experiences very much the ego goes away I'm very suspicious of the claim that the self goes away because people are readily able to recall these experiences this is what Forman does report and I would report to you and seamlessly integrate them into their autonotic autobiographical memory there isn't any weird dis jarring or where was I during yeah yeah yeah yeah yeah okay excellent the reason why I asked that is because um a lot of self modeling in active inferences based on the work of Thomas matzinger so Thomas matzinger is such a wonderful addition to this conversation so had a wonderful conversation with him not that long ago really love on my Chanel amazing I think he's I think he's such an underrated and important philosopher totally agree yeah yeah so he's got this notion of so he makes a distinction between phenomenal self modeling and an ontological self so people intuitively even everyone listening to this podcast even the way we act on a day-to-day basis is in Li of an ontological self so we can't help but really think of ourselves as being this Soul this Cari and soul but anyway Metson invokes this notion of phenomenal self modeling how the system appears to itself and he's got this um minimal phenomenal self which which has been unpicked in other ways but he normally he speaks generally of presentness mindness and perspec ible exactly what I've been talking about yeah exactly this gets fleshed out in some of the act of inference work people like jaob ho um Carl and yob lonowski have used um metsing good work for several papers and then on the other end of that we have what Carl turned an epistemic agent

model which is the self that sums the system that sees itself as epistemic in the sense that it can retrieve past memories to inform future decisions and agentive in the fact that it can conduct allostatic action. I.E. sum action to retain some homeostatic equilibrium given what it knows about the future so me putting on a jumper before I go outside because I know it's going to be colder outside than it is inside the reason why I invoke the minimal phenomenal selfhood is because and Carl actually outlines this in the podcast we did together that Consciousness may be Downstream in a sense on one a deep temporal model and from that the sense of agency and he has an argument from sum referring to Dimensions sum not in terms of sort of millimeter size but in terms of sum the degree to which you have embedded Markov blankets sum that when we're distant from our actuators what we end up doing is we have to have a way of distinguishing what's my action from what's your action or from what the world how the world has acted on us and in doing that we come up with a self other distinction but also the notion of an agentive self the idea that I can change the world in accordance to preferences as I mentioned beforehand so to sort of paraphrase Carl what that seems to be suggesting is that Consciousness is actually rooted in selfhood and I know there's this whole argument in Metzinger about whether that's the case that's why I ask whether mindness is also part of these pure Consciousness events or whether it's kind of a post talk inference just wondering whether you had any ideas about that notion sum and whether we can truly have have Consciousness without at least a sense of self right okay so sum uh let me try a few things first of all sum I don't think it's inevitable that human beings model themselves as Souls I think that's a western post Cartesian uh way of and that's even and then the Soul as a uh monadic single substance can put it that way uh you don't that's not even the case in uh the ancient world of the West certainly not the case in other parts of the world Etc so I I I I hesitate that that's sum the claim that's part of how the Machinery must unfold sure and then secondly sum I I worry about saying you know that that that because it isn't a soul there isn't a self this is a weird notion of again of a substance ontology being just presupposed in an unquestioned manner I mean look we've discovered most things aren't substances this table which would be a classic arisan substance is not a substance it's it's a dynamical system of atoms and quarks and blah and we don't go oh well because tables aren't substances they aren't real like that we don't do that so I like I I I I I just I I want to just note that I'm worried that there's a substance sure ontology creeping in here and leading to certain conclusions now that idea about uh uh this there's I can't remember the name of the people uh and I apologize for that there is a fairly recent Theory Of Consciousness that trying to respond to all the liit experiments and things like that

um arguing that well conscious ious is sort of after the fact but Consciousness is actually for the future yeah conscious so the idea here is consciousness is emerges out of the evolution of episodic memory so the function of Consciousness is to allow us to create an episodic memory of something we have already done uh right and the point about the episodic memory is soon as you get into episodic memory you get into perspectival knowing that's what an episode is right you have a perspective right on a situation what you found Salient and relevant how your actions and right and the arena coupled or didn't couple the affect that all all of that all of that that's so bound up with Consciousness and and then the and the point that they make and I totally agree with this is soon as you agree that there are multiple kinds of knowing not just propositional and procedural but also perspectival and participatory you get the argument that episodic memory affords perspectival knowing which allows you to solve problems that you can't solve without perspectival knowing you can pick up on the world the world discloses itself in ways that it is not otherwise disclosable to you and then Consciousness emerges as an optimization on the formation of episodic memory so it is optimally transfer appropriate for the future right and then you get it is a sense of agency but it's not a billiard ball agency yeah it's the it's this kind of longitudinal agency and to tell you the truth given the humi and critiques that's actually the kind of agency the self has it has this kind of longitudinal uh right agency around episodes it doesn't have this I'm an some sort of uncaused cause and moved mover Within Myself which I think is both a ridiculous proposal ontologically and an and an ethically undesir why would I want such a thing right like it's completely arbitrary I I think my life is being lived I'm trying to change myself so that I am my thoughts as determined by what is true my action as determined by what is good and my uh perception is determined by by what is beautiful as I possibly can I would like to lose all my freedom in that sense um um and so I think if we move to to the right level of analysis and Gallagher makes a convergent argument about this that when we're talking about agency and selfhood we're not talking at that liit scale we're not talking at where's the first movement of the billiard ball chain we're talking more about no no no how are we building this longterm virtual engine that enhances our predictive agency in the world yeah absolutely and that and Gallagher's work is very convergent with Thomas matzinger of course uh and so this just for our audience this these arguments and this notion of the narrativized self is well fleshed out in active inference literature um so I could point you yeah I would probably start with the the fris and lonowski papers there are two wonderful papers about self-construction under Act of inference and and then I have a convergent argument coming out of Daniel Daniel hudo who is also in the 4E

right the narrative practice hypothesis and he argues that any mindsight ability any ability to see in other people's into other people's mental States you know attributing beliefs and desires to them uh requires well for example if I'm going to attribute a belief and desire I need to know something about your character yeah are you lying or not right I need to know something about the context the setting like what's going on in this situation are you tired is that a small child so you're not really lying when you tell them at Christmas time that there's a Santa Claus right uh right and I need to know what the is what the problem notice what I'm talking about here I need to know all the elements of narrative Daniel Judo points out we practice narrative incessantly and unlike many of the other things including language which we scaffold for children we scaffold narrative for our our kids I had to sit through the Teletubbies twice in which you have to sit through these really impoverished narratives because we're scaffolding this up because narrative ability gives us the right gives us the set of skills the sets of states of mind with the perspectives taking therein and the traits of character that allow us to pick up on other people's mental States and I think that is also a function of the S the function of selves is to make us agentially predictive to each other right exactly exactly and there's another there's another convergent argument in Act of inference which is that um only in the context of other people like you would you ever come to the inference that there is something that is like to be you yeah yeah if you were cones with vski that converges with the whole vatan approach to the development of metacognition and self-awareness which I think is right right right right and of course um your colleague at um just pointing people in different directions your colleague at Toronto or former colleague Jord Peterson um yeah has a whole uh literature on literature and narrative and he has a conversation with car on his podcast which is um deeply intriguing about these kind of alignments of the free energy principle and and narrative John I wanted to also speak about flow States um you've written about flow as yes the sort of locus of implicit learning um as you know I've just been writing up a paper on Flow um from an active inference perspective so I'd love to be able to sort of see maybe where we align where we don't align and try and sort of unpick that just as a for warning my paper the paper that I've written with my wonderful co-authors is not out yet but it should be coming out soon um so I want to I have not yet read your paper because you you suggested waiting uh yes the paper paper's gone through um modifications as papers are want to do it's in the final stages so you have access to it so so please feel free to read it I will read it then I will read it going for I can give I can give a very brief overview so to not disadvantage you um my the basic thesis of the basic perspective that we take on the paper is about self- modeling under

flow States so that's the main thing that we're looking at and the attenuation of an epistemic agent model because what we're arguing is that certain Precision weighting mechanisms are leading it are leading the organism to undertake pragmatic action seeking out pragmatic affordances um rather than engage in what car or other offers would call epistemic foraging now I think where we might have a point of difference here is that I have a section on Flow States and learning and I think this is a really interesting point because your paper speaks about implicit learning whereas I'm and and this and to to make it transparent this is there are I've heard different perspectives on this but my current opinion is that in flow States what you're getting is a reinforcement of the skills that you actually um picked up through epistemic work so I have the example of a violinist the violinist must undergo a certain period of exploration of epistemic foraging which comes with this kind of self talk this real prominence of myself as a knowing thing yeah right and then what it does is over time through learning you get high Precision um over the beliefs about that action in terms of the technical detail and then that Bec becomes kind of the foundation on which you can then go and do more epistemic work and then that become and then that expertise development is stepwise in a sense I mean you can picture it either way but critically and where we differ is that I'm arguing that actually in the moment of flow itself you're just garnering more evidence for the capacities and the policies that you already have so I'd like to start there and see and see what you think of that claim um so we as you point out Leo Ferraro AR and her and I we argue um something different um which is why you're bringing it up uh we argue that um um that learning is inevitably occurring even in the flow situation and so there's Improvement um and that's why if the environment doesn't have the capacity to renew uh it challenges on you you will very quickly fall out of the Flow State so the argument is well why don't you just stay in the Flow State well the argument is because eventually you get a Mastery over the environment um which means you still start to exceed the demands and so phenomenologically when I'm in the Flow State like in sparring or lecturing um I'm finding tremendous amount of insight and Innovation uh coming out um now this is where it might get tricky um because is procedural Innovation a restructuring especially uh is that a restructuring of um uh your information and therefore have new capacities in it new emergent abilities or is it just a reinforcement and I don't this we might get into thesis's ship here which is going be problematic uh but the one of the defining phenomenological features of the Flow State and I'm uh I want to be clear I'm not pinning you down on this but I do think it's relevant evidence is the ongoing sense of Discovery there's a sense of Discovery there there's a sense of coming

to know things you did not know before and of course when the flow state is in much more comprehensive expertise not like they playing tennis or maybe like just your optimal gripping on the world people come to think they have learned something deeply profound uh about reality so I tend to think that there's evidence for transformative learning happening up and Beyond just reinforcement Lear okay yeah I I I like the return to the phenomena um you know draas has this phrase which I will definitely use over and over again in this podcast which is when in doubt return to the phenomena um the way that we have cast it is that there's this idea in the literature about a hyper prior that the world is changing so that we have a prior that the world is changing and that is adaptive for us because we don't get stuck in the same free energy Minima now if that prior is always at play what ends up happening Downstream on that is that my inference about my own abilities deteriorates over time and we and we we experience that I guess I mean like if I play tennis today and then I play tennis in a week's time I'm probably going to have less Fidelity in my own capacities in the week's time than I do if I play tomorrow having follow today yeah yeah and so our argument is that the positive effect that's part of the flow state is Downstream on the rise that's generated when you actually violate that hyper prior that the world is changing because what you're getting evidence for is that your policy still work in the world that you would have beener you would have thought didn't lend itself to your policies working and there's this idea um Casper hp's paper and his co-authors in 2021 that affecting active inference is about um basically your model doing better than you thought it would do um and and this actually comes a lot into what Mark talks about in terms of aerodynamics so so I'm wondering yeah I don't know yourself well see I I think that's right I think and and the problem of course is you can get an infinite regress so well I'll say well the mod the the things here and you'll say well there needs to be something a h a hyper prior behind that and then we can that's what I mean about thesis's ship yeah yeah yeah so again I would say well what what are you getting better at and what you're getting better at I would argue is um not St like so expertise is generally built around like uh giving you a well like a specific domain we can talk about um the possibility of sort of meta expertise if we want but what I mean by that is once you've got expertise you've gotten really good at within a certain domain at formulating the problems you're fronting as for you well- defined problems uh that's one way in which an expert is sort of reliably different from a novice a novice goes in this is an IL defined problem the expert goes goes in and say no it's this problem and this is the right what you need to do this and this and this and this okay so we agree with that but I think what's happening right and and maybe this is the gray area

between your position in mind I think what's happening is we're discovering new ways in which we can turn well-defined problem sorry ill-defined States into well-defined problems we're going something like I didn't realize I could adapt to that situation but I can yeah and and that sounds similar uh but for me that's that's an Insight experience uh and that's what like it's an Insight flow because you're you're you're getting an extension of your cognitive capacity because you have restructured what you take to be the like your abil your your problem formulation of the world that's that's yeah I mean to supplement that I I actually I I don't disagree um on that not but the way I would formulate that and this actually isn't in the paper it's in the paper in some sense but it's not a fully fleshed out argument because it wasn't hyper relevant I'm glad there's another really quality paper on Flow coming there not there's really there's really nothing um when we published the paper we were literally the only paper talking about the cognitive processes that work in flow I I yeah it's it's a it's a bizarre one um there's very little on skillful coping generally within yeah Cog science but as I said I think that's why I said there should be more phenomenology active or cognitive science convergence what what we kind of mentioned in brief Is That Flow if we take the sort of macro perspective on what flow is it can also be this kind of humming between the pragmatic and the epistemic boundary yeah so maybe when you're realizing oh I can reframe this um kind of uh perplexing problem as actually something manageable what you're doing there is Maybe just slightly exiting the flow doing some epistemic foraging returning because something I want to make really clear to everyone listening as well the when we talk about epistemic action and pragmatic action in terms of expected free energy or active inference it's not like the agent goes I'm doing pragmatic action right now anding right it's all part of the actual maths right so if you looked at the maths it's like firstly you can do a bit of both right at the same time like that that's a thing but it's also like the action policies that we're talking about here John and I these are not protracted seven hours of just like pragmatic action no no these have to be very temporally thin because the world is volatile right the world is constantly changing and so you have to be able to not just pin all your hopes or put all your eggs in one basket in terms of an action policy and so actually fundamentally our Divergence might well be a semantic thing which is that I'm talking very much about the synchronic nature of being in flow right now and like I have for for Simplicity I have that boundary between pragmatic and epistemic whereas if you take a kind of more macro perspective as that flow State as the concerto unfolds you could be humming at that boundary and doing learning as well yeah I I I really look forward to your paper because I think that sounds like actually a powerful way in which the theories could be

integrated uh together um I'm also interested in a question that's emerging out of this dis well I I've already been interested in the question but it emerges out of what we're talking about because I'm interested because I think there's there might be an additional thing um because there's you can flow in a situation and it doesn't transfer it doesn't transfer to other domains video game addiction is the classic Model you can flow in the game you can't flow in the world so you get depressed in the world you can flow in the game and so you want to stay in the game more and more and more um that for that's very different than how I flow in taichian taichian in fact other people pointed this out in me the the the flow states that I cultivate in tai chi Chan I can they are cultivated in such a way that they transfer broadly and deeply to many other domains and I'm interested in the this this this this sort of ritual framework because what what you get is this ritual and this philosophical framework do some and other e other practices and ecologia practices around so that it broadens um like it transfers in powerful ways and and I'm wondering um what the differences are and also if how does that show up is there a phenomenological difference there I mean so now really really really weakly anecdotally right uh in terms of phenomenological practice I do get a sense of a difference when between when I'm just flowing and when I'm flowing in a ritual context so if I enter into a violin Hall and like let's say an expert in playing the violin and I see a crowd and I see my violin and I see the whole stage what I end up getting is this um so habits are quite well codified within Act of inference you get this kind of contextual cue that like in layman's T although this is we're saying this isn't happening propositionally it's time for me to play the violin and this can look like Precision over beliefs about your own action so what this is again not something I've thought about necessarily with any depth because is just something that you've brought up I'm wondering whether that contextual cue what you're happening when you're doing taichi and then maybe what you're doing when you're lecturing is that your your system is seeing similarities in that in the contexts and doing similar Precision waiting in other in in a separate do in a separate domain yeah I think that's right and I think what philosophical Frameworks do is they reverse engineer that they try to uh find out what might be similar or maybe even invariant across many contexts and then build that back into the specific context in which you are doing the practice to exactly afford that transfer appropriate processing I think that's right I think there's something else also going on with the queuing and this goes into apter's meta motivational theory we can be e we can be broadly speaking we frame our arousal in two different ways when we're in a telic mode and we're working towards some where where where the reward is found in the external goal then increased effort is

experienced as frustration and and then it and then it increases too much I get anxiety but I mean if I'm doing something where the goal is the behavior itself like making love or doing poetry then the increased not infinitely obviously but the increased arousal is framed as positive as excitement right right right right uh and so I think also and so after talks about safety uh safety frame right let's call the first atelic Mode work and the um the paratelic mode um um play and I think ritual has to do with serious play and I've got a whole argument about that but I think also what what you're doing in the context is you're you're trying to cue people into the play mode uh because the play mode allows them to model themselves their arousal as positive rather than as negative and that allows them and I think there's a deep connection between this paratelic Framing and chick sahi's I think accurate claim that flow is autotelic doing it for its own sake right yeah yeah cool yeah I I like the notion of effort here recognize that one of the fundamental factors or fundamental characteristics of flow is uh perceived effortlessness yes even though you can even though you can have sort of very uh peripheral cues that you might be expending a lot of metabolic energy exactly right um and just for the given this is the active inference institutes podcast just for those interested in the computational framing of that within active inference Thomas par has got a paper out this year on cognitive effort and active inference so that gives a kind of nice computational modeling picture of that which actually leads me to a much broader question because I know we're we haven't got too much more time which is I've noticed in myself to be totally candid that I've started like I I learned about flow and sixcess M High through you and and through other people but mainly as a philosophical notion um and now I'm viewing it as a computational notion yes yeah do you worry about that do you worry that if we rely too much on comput not not just computational models but maths physics we in some ways reduce the phenomena in a way that's detrimental not just because it's not romantic or that it's but like that we're missing something fundamentally well it depends I mean that's a really important question we could do two hours on just that question uh but if you think that a leveled ontology is actually the way ontology is um that the level at which we do science is as ontologically real as the quantum level we discover when we're doing science and I think you have to come to that conclusion right and I have extended Arguments for that elsewhere then uh there's then you can make a clean philosophical distinction between explaining a way in a reductive fashion and explaining that actually enriches your appreciation for the phenomena uh now if the computational stuff the computational stuff to my mind well let's say what the argument we were just exploring um has Merit then notice what we're saying we're saying well what the computational

modeling actually does is actually shows why this sometimes very obscure philosophical framework is really important yeah it's actually doing some really important work and you can't get rid of it you can't dispense with it and and and and and that would mean that kind of some of the stuff we're doing where we're trying to commodify flow and take it out of those those Frameworks you know and do the thing we do and you know sell books and right um it's actually Mis misrepresenting the phenomena in a in a in a way that we can philosophically and scientifically critique and say wait wait right this phenomena has the power it has because chick sahai said it's an evolutionary marker for adaptivity and then well making it adoptive is how broadly and deeply is it transferring out of the situation and then the philosophical framework really matters to what it is and how it functions and so if you if you can make a distinction and you need an ontological distinction to make this distinction but if you can make a distinction between explaining and explaining away then then I think it's possible to say no no when I do these things and and to be fair to me darus I get a lot of people people who you know are from various religious and spiritual backgrounds and they come in and they say thank you so much for your work I now I much better appreciate this right this experience or that experience or the Flow State or this mystical State because you didn't try and tell me it's nothing but you tried to say this is what it's doing and this is why you like it and value it so much but do you think your work to kind of yeah to to look at that personal vantage point do you think you've managed to keep like cuz if I without being sort of embarrassingly lording people should watch Awakening from the meaning crisis um not only for the content but also just the way you present it which is just so magical um so thanks I mean it's it's really inspired a lot of the way that I think science should be done and philosophy should be done what what is it about I think you know anyone who watches that recognizes there's something kind of special going on there the ideas are living through you you're they're breathing through you you're breathing through them there's this really beautiful coupling again we'll come back to that between you and the ideas what what is it about that kind of symphysis of the philosophy and the sort of more hard cognitive science do you think made that project so successful I suppose it's a view of the role of cognitive science what cognitive science is doing um I think what cognitive science does it well I'll try and do it sort of narratively uh uh and I do this in the series I think you know the even the notion of Mind we've been invoking mind and cognition is equivocal uh because it means one thing to the neuroscientists who's studying the brain and looking at uh neurons and anatomical networks and perhaps functional networks it means a different thing to the artificial intelligence person who's building algorithms and

puristic um and doing reinforcement learning and blah blah means a different thing to the psychologist who's studying human behavior with experiments and running stats and they talk about working memory they don't talk so much right and it means a different thing to the linguist and a different thing to the cultural Anthropologist by the way I include them because they were the people that have been studying distributed cognition and collective intelligence they mattered yeah um now if you think about it on this analogy it's like there are different countries speaking different languages and here's the thing that has great value specialization has great value I'm not dismissing that I am not dismissing that but each one of these is talking about a different level and here's what I think I would State as a very plausible claim I think it's unlikely that these levels in reality the brain information processing behavioral linguistic sociocultural levels are independent from each other I think they causal influence and constrain each other so we are missing something important about the Mind by right not getting clear about the relationship between these levels we can be equivocating and we are we have a fragmented notion if we don't capture the relations of constraint and causation between these different levels of cognition and mind and so I think the proper function of cognitive science is to use philosophy's skills of creating bridging discourses creating bridging conceptual right uh vocabulary theoretical grammar so that these different disciplines can talk to each other in reciprocally reconstructive ways as a way of converging on the causal and constraint relationships between the levels rather than trying to compete or say the bottom level is the only real real level so that for me they all live together they are doing and I'm sorry to invoke it again they're doing opponent processing between each other U and that's how that's that's what that's that's what inhabits me when I'm doing cognitive science that Vision I call it synoptic integration yeah it's well it's wonderful to see um so so I mean like I think we I mean that's the cognitive scientist that I'm aspiring to be um I think you've called it a big picture cognitive scientist as well yes and one of the things one of I felt very only for a while you know I I was doing relevance realization which is a big picture and then to my great Delight another big picture for EOG side down the and then to my even greater happiness another big picture came down the road which is you know the predictive processing framework and I think they are more convergent than op uh adversarial and I well you've heard me arguing about how we can integrate them together excellent yeah yeah yeah I would say maybe it's just bias active inferences or prics processing is a big picture cognitive scientist from the physics and the maths all the way what what we're trying to do to the phenomena to what it is like to be a to be a human being to have conversations like this John it's um I it was

you know I can't say it was better than I expected because I knew it was going to be wonderful but um you're full of just thrilling insights and speaking to you listening to you is always such a wonderful learning experience where that happens in flow or without flow um but I just wanted to thank you so very much for I know you're extremely busy um but for giving us your time where can people find you what have you got coming up um I'm sure people are curious so the most immediate thing for this conversation um is the talk I gave at Leiden on the predictive processing Symposium it's up on my channel I think it's the second most recent video on my channel on YouTube we can put it in the yeah where I I I try to go into the nuts and bolts of how you could integrate uh predictive processing and relevance realization Theory together so people might find that and I've got a lot of good feedback on that uh for being sort of clear clear and and a good argument um so I would recommend that when people are interested in uh the the the the broader implications of this um Awakening for the meeting crisis and then after Socrates also after Socrates is where I take all of this in all of this stuff and well how do you turn it into practices in order to overcome self-deception and enhance relevance realization become more wise and virtuous um and so they can take a look there um the arguments around neoplatonism um there are several videos around that how I've tried to connect neoplatonism to for eoai and relevance realization of predictive processing as as we saw in this podcast but in the neoplatonism I'm working on my third big series um walking the philosophical Silk Road which will be on Zen neoplatonism trying to see if we could bring an integration an opponent processing not an adversarial one an opponent processing between Zen and neoplatonism to give us a rich philosophical framework by which like the philosophical road we could trade ideas and move between worlds without um having to descend into well tribalism and other such things wonderful wonderful well again it was absolutely my pleasure I apologize I've got a slight cold so if I've been a little sniffly or nasle oh no we blame on we can blame it on the London weather but um I'd love to have you back on at some point um your your work is is truly inspiring so thank you so much John for me and The Institute thank you darus I'm happy to come back on and you know if it if it turns out that you know Mark and I come on together that would be thrilling as well great we will definitely get that sorted all right thank you thank you